

Mathematics for AI - June 2026 - PS1

Question 1

Translate the following sentences into propositional logic. Your formalizations should be as detailed as possible.

- a) Three is a prime number and is not an even number

Answer:

Key:

P = Three is a prime number

Q = Three is an even number

$$P \wedge \neg Q$$

- b) "If the speed limit is 30 mph and I am driving at 25 mph, then I am not breaking the law."

Answer:

Key:

P = the speed limit is 30 mph

Q = I am driving at 25 mph

R = I am breaking the law

$$(P \wedge Q) \rightarrow \neg R$$

- c) "Bob will go to the grocery shop only if the fridge is empty."

Answer:

Key:

P = Bob will go to the grocery shop

Q = the fridge is empty

$$P \rightarrow Q$$

Question 2

Translate the following sentences into predicate logic; formalizations should be as detailed as possible.

- a) Fairies do not exist.

Answer:

Key:

F^1 = ...is a fairy

$$\neg \exists x Fx$$

- b) Somebody has visited all the concert halls in London.

Answer:

Key:

P^1 = ...is a person

Q^1 = ...is a concert hall in London

R^2 = ...has visited...

$\exists x Px$ there exists something such that they are a person (somebody)

$\forall y (Qy \rightarrow Rxy)$ looking across all y , if y is a concert hall in London then x has visited it.

Putting it all together:

$$\exists x (Px \wedge \forall y (Qy \rightarrow Rxy))$$

- c) Not everyone has a sibling.

Answer:

Key:

S^1 = ...has a sibling

$$\neg \forall x Sx$$

Question 3

Consider the propositional formula

$$F = ((R \wedge I) \vee (\neg S \vee \neg P)) \rightarrow \neg C.$$

with R = “the router is on”, I = “internet is available”, S = “signal is strong”, P = “entered password is correct”, C = “the device is connected”.

- a) Translate F into an English sentence.

Answer:

F = “If either the router is on and the internet is available, or the signal is not strong or the password entered is not correct, then the device is not connected.”

- b) Which situation is modelled by that sentence, under the interpretation above?

Answer:

Having read about the contrapositive being logically the same as the original statement (Grossman, 2009, pp.54-55), I knew it would be a helpful tool here. This question asks what situation this describes, but the sentence is written in the negative form (in terms of what it is not), which is not the most natural way to explain and answer this question.

If we write the contrapositive:

$$C \rightarrow ((\neg R \vee \neg I) \wedge (S \wedge P))$$

So using the contrapositive it is: “if the device is connected, then either the router is not on or the internet is not available and the signal is strong and the entered password is correct.”

In conclusion the situation being modelled is the case where the device is connected only if the signal is strong and the password that was entered is correct and either the router is off or the internet is down. This is how the logic pans out but it does seem odd to me compared to how devices usually work.

- c) Assume R, I, C, S are true and P is false. Is F true or false? Briefly explain why.

Answer:

$$F = \text{False}$$

Reasoning: If we stick with our contrapositive expression:

$$C \rightarrow ((\neg R \vee \neg I) \wedge (S \wedge P))$$

Considering True as T and False as F, we get the following expression:

$$\begin{aligned} T &\rightarrow ((F \vee F) \wedge (T \wedge F)) \\ \therefore T &\rightarrow (F \wedge F) \\ \therefore T &\rightarrow F \end{aligned}$$

So F is **False**. If we consider each proposition’s truth value and then evaluate those with the connectives for the antecedent and consequent we get $T \rightarrow F$. We know from the truth table for if-then this is False so the final answer is False (Grossman, 2009, p.49). Deciding this is purely based on the atomic proposition’s truth values and does not have anything to do with causation from the antecedent to the consequent.

Question 4

The sentence “Every device that can reach a router, and knows the password of that router, will connect to it” is formalized as

$$P = \forall d \forall r ((\text{Can-reach}(d, r) \wedge \text{Knows-pwd}(d, r)) \rightarrow \text{Connected}(d, r)).$$

Consider also

$$Q = \forall d (\exists r (\text{Can-reach}(d, r) \wedge \text{Knows-pwd}(d, r)) \rightarrow \exists r \text{ Connected}(d, r)).$$

1. In a domain with 2 devices and 3 routers, give a model in which P is falsified.

Answer:

P contains a conditional which is false if the antecedent is true but the consequent is false (Grossman, 2009, p.49). P uses universal quantifiers for d and r so we only need one pair of elements from the domain where the conditional is false for P to then be false in our model.

Below I provide the model $\mathfrak{A}$ for domain $|\mathfrak{A}|$, such that each predicate B gets an n -ary relation: $B^{\mathfrak{A}} \subseteq |\mathfrak{A}|^n$, (Enderton, 2001, pp.80-81).

Model $\mathfrak{A}$:

$$\text{Domain} = |\mathfrak{A}| = \{d_1, d_2, r_1, r_2, r_3\}$$

$$\text{Can-reach}^{\mathfrak{A}} = \{(d_1, r_1)\}$$

$$\text{Knows-pwd}^{\mathfrak{A}} = \{(d_1, r_1)\}$$

$$\text{Connected}^{\mathfrak{A}} = \emptyset$$

If we consider $d = d_1$ and $r = r_1$, where d_1 can reach r_1 and it knows the password to r_1 but it is not connected to r_1 , then we have $(\text{true} \wedge \text{true}) \rightarrow \text{false}$ which is false, and from the reasoning given above, P is false in this model $\mathfrak{A}$.

2. Explain whether Q provides a different formalization of the original sentence than P , and why.

Answer:

Q does provide a different formalization of the original sentence, compared to P . In Q the $\forall r$ has been replaced by two $\exists r$, one in the antecedent and one in the consequent. P states that if a device can reach and knows the password of a router it should connect to that router. Q says it just connects to some router, it does not have to be the same router. Consider the below model $\mathfrak{A}$:

Model $\mathfrak{A}$:

$$\text{Domain} = |\mathfrak{A}| = \{d_1, d_2, r_1, r_2, r_3\}$$

$$\text{Can-reach}^{\mathfrak{A}} = \{(d_1, r_1)\}$$

$$\text{Knows-pwd}^{\mathfrak{A}} = \{(d_1, r_1)\}$$

$$\text{Connected}^{\mathfrak{A}} = \{(d_1, r_2)\}$$

P is false due to $(d_1, r_1) \notin \text{Connected}^{\mathfrak{A}}$ and because of the universal quantifier on r the same r_1 from the antecedent needs to be in the consequent. Q is true as the r in the antecedent can be different to the one in consequent. Since Q is true but P is false in the same model, they are not the same formalization of the sentence.

Question 5

For each relation, state clearly which of reflexive / symmetric / transitive hold, and whether it is an equivalence relation. (No working required.)

1. A = set of all animals, $R_1 = \{(a, b) \mid a \text{ is the same species as } b\} \subseteq A \times A$.

Answer:

Reflexive, symmetric, transitive and $\therefore$ equivalence relation

2. $R_2 = \{(m, n) \mid m^3 \leq n^3\} \subseteq \mathbb{Z} \times \mathbb{Z}$.

Answer:

Reflexive, transitive but since it is not symmetric it is not an equivalence relation.

3. $R_3 = \{(x, y) \mid x + y < 1\} \subseteq \mathbb{R} \times \mathbb{R}$.

Answer:

Symmetric, since it is not reflexive or transitive it is not an equivalence relation.

4. $S = \{a, b, c\}$, $R_4 = \{(a, b), (b, c), (c, a)\} \subseteq S \times S$.

Answer:

Not reflexive, not symmetric and not transitive so it is not an equivalence relation.

Question 6

Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be defined by

$$f(x) = \begin{cases} \frac{x}{2}, & \text{if } x \text{ is even,} \\ -\frac{x+1}{2}, & \text{if } x \text{ is odd.} \end{cases}$$

1. Demonstrate that f is injective.

Answer:

Proof. We need to show that f is injective so for every output in the codomain there is at most one input from the domain i.e. for every $x_1, x_2 \in \mathbb{N}$ if $f(x_1) = f(x_2)$, then $x_1 = x_2$:

The domain is the Natural numbers $\mathbb{N}$ which throughout this question are considered to be $\{0, 1, 2, \dots\}$ so they can only ever be even or odd. There are then three possible scenarios to consider. If we have $x_1, x_2 \in \mathbb{N}$ and assume $f(x_1) = f(x_2)$:

Scenario 1 where both x_1 and x_2 are even:

$$f(x_1) = f(x_2) \Rightarrow \frac{x_1}{2} = \frac{x_2}{2} \Rightarrow x_1 = x_2$$

Scenario 2 where both x_1 and x_2 are odd:

$$f(x_1) = f(x_2) \Rightarrow -\frac{x_1+1}{2} = -\frac{x_2+1}{2} \Rightarrow x_1 = x_2$$

Scenario 3 where one of the x values is even and the other is odd:

Consider $x \in \mathbb{N}$. If x is even, then $x \geq 0$ so:

$$f(x) = \frac{x}{2} \geq 0$$

If x is odd, then $x \geq 1$ and $(x+1) \geq 2$ so:

$$f(x) = -\frac{x+1}{2} \leq -1$$

So every even input has a non-negative output and every odd input has a negative output. Since one of x_1, x_2 is odd and the other is even, it is not possible for them to produce the same output so our assumption is violated, $f(x_1) \neq f(x_2)$, so **Scenario 3** is not possible.

In all of the possible scenarios where $f(x_1) = f(x_2)$, $x_1 = x_2$ so f is injective. $\square$

2. Prove that f is a bijection.

Answer:

Proof. We need to show that f is a bijection. By definition a function is a bijection if it is injective and surjective. We have already proven it is injective in question 6.1 so we now just need to prove surjectivity. A function is surjective if for every element of the codomain $y \in \mathbb{Z}$ there exists some element of the domain $x \in \mathbb{N}$ such that $f(x) = y$.

The codomain is $\mathbb{Z}$ meaning its elements can have $y \geq 0$ or $y < 0$. That gives two possible scenarios to consider. If we have $y \in \mathbb{Z}$:

Scenario 1 where $y \geq 0$:

$$x = 2y \Rightarrow x \geq 0$$

so $x \in \mathbb{N}$ and x is two times an integer so it is even. Since x is even:

$$f(x) = \frac{x}{2} = \frac{2y}{2} = y$$

Scenario 2 where $y < 0$ which since $y \in \mathbb{Z}$, $y \leq -1$:

$$x = -2y - 1 \text{ and since } y \leq -1 \text{ then } -2y \geq 2 \Rightarrow x \geq 1$$

so $x \in \mathbb{N}$, and since $y \leq -1$, $-2y$ is a positive even integer, so $x = -2y - 1$ is odd. Since x is odd:

$$f(x) = -\frac{x+1}{2} = -\frac{-2y-1+1}{2} = y$$

Since in both scenarios for every $y \in \mathbb{Z}$, we provided an $x \in \mathbb{N}$ such that $f(x) = y$, then f is surjective. Since f is surjective and injective it is therefore bijective. $\square$

3. Find the inverse function $f^{-1} : \mathbb{Z} \rightarrow \mathbb{N}$.

Answer:

The inverse is where the domain of f which is $\mathbb{N}$ becomes the codomain of f^{-1} and the codomain of f which is $\mathbb{Z}$ becomes the domain of f^{-1} .

We proved in question 6.2 that f is a bijection so for every $y \in \mathbb{Z}$ there is exactly one corresponding $x \in \mathbb{N}$ with $f(x) = y$. From question 6.1 scenario 3, when x is even, $y \geq 0$ and when x is odd, $y < 0$ and it is these conditions for y that give us our conditions for the inverse.

When x is even and $y \geq 0$, $y = \frac{x}{2}$. Solving this for x gives $x = 2y$. When x is odd and $y < 0$, $y = -\frac{x+1}{2}$. Solving this for x gives $x = -2y - 1$.

So $f^{-1} : \mathbb{Z} \rightarrow \mathbb{N}$ is defined by:

$$f^{-1}(y) = \begin{cases} 2y, & \text{if } y \geq 0, \\ -2y - 1, & \text{if } y < 0. \end{cases}$$

Question 7

Consider $f, g : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = \sin(5x + \pi), \quad g(x) = \sin(2x)e^{2x}.$$

Show some working before each final answer.

1. Find $f'(x)$ and $g''(3)$.

For $f'(x)$ we need to apply the chain rule. Let the following be defined as:

$$\begin{aligned} u(t) &= \sin(t), \\ v(x) &= 5x + \pi, \\ u'(t) &= \cos(t), \\ v'(x) &= 5, \end{aligned}$$

So,

$$f'(x) = \cos(5x + \pi) \cdot 5 = 5 \cos(5x + \pi)$$

For $g''(3)$ Let the following be defined as:

$$\begin{aligned} q(x) &= \sin(2x), \\ u(t) &= \sin(t), \\ v(x) &= 2x, \\ u'(t) &= \cos(t), \\ v'(x) &= 2, \\ p(x) &= e^{2x}, \\ p'(x) &= 2e^{2x} \end{aligned}$$

Applying the chain rule for $q(x)$ we get:

$$\begin{aligned} q'(x) &= \cos(2x) \cdot 2 = 2 \cos(2x) \\ q''(x) &= 2 \cdot (-\sin(2x)) \cdot 2 = -4 \sin(2x) \end{aligned}$$

Now we can apply the product rule:

$$\begin{aligned} g'(x) &= q'(x) \cdot p(x) + q(x) \cdot p'(x) \\ &= 2 \cos(2x) \cdot e^{2x} + \sin(2x) \cdot 2e^{2x} \\ &= 2 \cos(2x)e^{2x} + 2 \sin(2x)e^{2x} \end{aligned}$$

For $g''(x)$ we need to use both product and chain rules:

$$g''(x) = -4 \sin(2x) \cdot e^{2x} + 2 \cos(2x) \cdot 2e^{2x} + 2 \cos(2x) \cdot 2 \cdot e^{2x} + 2 \sin(2x) \cdot 2e^{2x}$$

$$\begin{aligned}
&= -4 \sin(2x)e^{2x} + 4 \cos(2x)e^{2x} + 4 \cos(2x)e^{2x} + 4 \sin(2x)e^{2x} \\
&= 8 \cos(2x)e^{2x}
\end{aligned}$$

So $g''(3)$ rounded to 3 decimal places:

$$\begin{aligned}
g''(3) &= 8 \cdot \cos(2(3)) \cdot e^{2(3)} \\
&\approx 3098.883
\end{aligned}$$

2. Calculate $\int f(x) dx$ and $\int_0^{\pi/4} g(x) dx$.

Answer:

Relying on the standard integral $\int \sin(ax) dx = -\frac{1}{a} \cos(ax) + c$:

$$\begin{aligned}
\int f(x) dx &= \int (\sin(5x + \pi)) dx \\
&= -\frac{1}{5} \cos(5x + \pi) + c
\end{aligned}$$

To integrate $g(x)$ we need to use integration by parts. Let:

$$q(x) = \sin(2x)$$

$$u'(x) = e^{2x}$$

$$q'(x) = 2 \cos(2x)$$

$$u(x) = \frac{1}{2}e^{2x}$$

$$\begin{aligned}
\int q(x) \cdot u'(x) dx &= q(x) \cdot u(x) - \int q'(x) \cdot u(x) dx \\
\int \sin(2x) \cdot e^{2x} dx &= \sin(2x) \cdot \frac{1}{2}e^{2x} - \int \left(2 \cos(2x) \cdot \frac{1}{2}e^{2x}\right) dx \\
&\text{to simplify let } A = \int (\sin(2x) \cdot e^{2x}) dx \text{ and } B = \int \left(2 \cos(2x) \cdot \frac{1}{2}e^{2x}\right) dx \\
A &= \sin(2x) \cdot \frac{1}{2}e^{2x} - B
\end{aligned}$$

We need to do integration by parts on B :

$$\begin{aligned}
B &= \int (2 \cos 2x \cdot \frac{1}{2}e^{2x}) dx \\
&= 2 \cos 2x \cdot \frac{1}{2} \cdot \frac{1}{2}e^{2x} - 2 \int ((-2 \sin 2x) \cdot \frac{1}{2} \cdot \frac{1}{2}e^{2x}) dx \\
&= \frac{2}{4} \cos(2x)e^{2x} + \frac{4}{4} \int ((\sin 2x)e^{2x}) dx \\
&= \frac{1}{2} \cos(2x)e^{2x} + \int ((\sin 2x)e^{2x}) dx
\end{aligned}$$

Now we can substitute B back into A :

$$\begin{aligned}
A &= \sin(2x) \cdot \frac{1}{2}e^{2x} - B \\
A &= \sin(2x) \cdot \frac{1}{2}e^{2x} - \left(\frac{1}{2} \cos(2x)e^{2x} + \int ((\sin 2x)e^{2x}) dx\right) \\
A &= \sin(2x) \cdot \frac{1}{2}e^{2x} - \left(\frac{1}{2} \cos(2x)e^{2x} + A\right) \\
2A &= \sin(2x) \cdot \frac{1}{2}e^{2x} - \frac{1}{2} \cos(2x)e^{2x} \\
A &= \frac{1}{2} \sin(2x) \frac{1}{2}e^{2x} - \frac{1}{2} \frac{1}{2} \cos(2x)e^{2x} \\
\int (\sin(2x) \cdot e^{2x}) dx &= \frac{1}{4}e^{2x}(\sin(2x) - \cos 2x) + c
\end{aligned}$$

Now we can compute the definite integral:

$$\int_0^{\pi/4} (\sin(2x) \cdot e^{2x}) dx = \left[\frac{1}{4}e^{2x}(\sin(2x) - \cos 2x) \right]_0^{\pi/4}$$

$$\begin{aligned}
&= \left(\frac{1}{4}e^{(\pi/2)}(\sin(\pi/2) - \cos(\pi/2)) - \frac{1}{4}e^0(\sin(0) - \cos(0))\right) \\
&= \left(\frac{1}{4}e^{(\pi/2)}(1 - 0) - \frac{1}{4}e^0(0 - 1)\right) \\
&= \frac{e^{(\pi/2)} + 1}{4} \\
&\approx 1.453
\end{aligned}$$

3. Is g differentiable at every point of $\mathbb{R}$? Justify your reasoning.

Answer:

$g(x) = \sin(2x)e^{2x}$ is differentiable for all points of $\mathbb{R}$.

First I inspected $g(x)$. There is no division that could bring about division by 0 issues; $g(x)$ is not a piecewise defined function, so there is no join where one-sided limits would disagree; and there are no even roots, so no risk of taking the root of a negative value.

Next apply the standard differentiation theorems to g . $\sin$ and $\exp$ are differentiable on $\mathbb{R}$ and $2x$ is a polynomial which is also differentiable on $\mathbb{R}$. $\sin(2x)$ and e^{2x} are made up of functions that are differentiable for all $x \in \mathbb{R}$ so by the chain rule these are differentiable on $\mathbb{R}$ too. $\sin(2x) \cdot e^{2x}$ is then a product of functions that are all differentiable on $\mathbb{R}$ so it is differentiable by the product rule for all $x \in \mathbb{R}$, to give:

$$g'(x) = 2\cos(2x)e^{2x} + 2\sin(2x)e^{2x}$$

4. What is the behaviour of $g(x)$ as $x \rightarrow \infty$?

Answer:

g is made up of $\sin(2x)$ and e^{2x} . I know $\sin(2x)$ oscillates between values of 1 and -1 . e^{2x} is always positive and keeps growing. For the maxima of $\sin(2x)$, which will continue to occur as x increases, it is 1 so $g(x) = e^{2x}$ which tends to $+\infty$. The minima also continue to occur as x grows, where $\sin(2x) = -1$, so $g(x) = -e^{2x}$ tends to $-\infty$.

Since g tends to $+\infty$ along the maxima and $-\infty$ along the minima, $\lim_{x \rightarrow \infty} g(x)$ does not exist and g will then just continue to oscillate, with an amplitude of e^{2x} as x grows.

Question 8

A square $n \times n$ matrix $M = (m_{ij})$ is *upper triangular* if $m_{ij} = 0$ whenever $i > j$ (lower triangular is defined analogously). The transpose M^T rewrites the rows of M as columns. Take

$$\mathbf{A} = \begin{pmatrix} 5 & 1 & -3 \\ 0 & 2 & 1 \\ 0 & 0 & -1 \end{pmatrix}.$$

1. Formally prove that for an *upper triangular* matrix M , the transpose M^T is *lower triangular*.

Answer:

Proof. If M^T is the transpose of M then $m_{ij} = (M^T)_{ji}$.

Consider any element m_{ij} where $i > j$. Since M is an *upper triangular* matrix, $m_{ij} = 0$.

M^T is the transpose, so the column index j becomes the row index for M^T and the row index i becomes its column index. So $(M^T)_{ji} = m_{ij} = 0$ when $j < i$.

Therefore every element $(M^T)_{ji}$ where $j < i$ is equal to 0, which is the definition of a *lower triangular* matrix. Thus M^T is *lower triangular*. $\square$

2. Compute $\mathbf{A}^{-1}$, its eigenvalues and eigenvectors; show some processing before the final result.

Answer:

To find $\mathbf{A}^{-1}$ I will use Gauss-Jordan elimination where I will row-reduce the augmented matrix $(\mathbf{A}|\mathbf{I})$ to $(\mathbf{I}|\mathbf{A}^{-1})$ (Singh, 2015, pp.110-112).

$$\left(\begin{array}{ccc|ccc} 5 & 1 & -3 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3 \rightarrow -R_3} \left(\begin{array}{ccc|ccc} 5 & 1 & -3 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\xrightarrow[R_1 \rightarrow R_1 + 3R_3]{R_2 \rightarrow R_2 - R_3} \left(\begin{array}{ccc|ccc} 5 & 1 & 0 & 1 & 0 & -3 \\ 0 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 5 & 1 & 0 & 1 & 0 & -3 \\ 0 & 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left(\begin{array}{ccc|ccc} 5 & 0 & 0 & 1 & -\frac{1}{2} & -\frac{7}{2} \\ 0 & 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\xrightarrow{R_1 \rightarrow \frac{1}{5}R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & -\frac{1}{10} & -\frac{7}{10} \\ 0 & 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \right)$$

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{5} & -\frac{1}{10} & -\frac{7}{10} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{pmatrix}$$

To find the Eigenvalues of $\mathbf{A}$ we need to solve the equation $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$

$$\mathbf{A} - \lambda \mathbf{I} = \begin{pmatrix} 5 & 1 & -3 \\ 0 & 2 & 1 \\ 0 & 0 & -1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} 5-\lambda & 1 & -3 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & -1-\lambda \end{pmatrix}$$

It is most efficient to evaluate the determinant along the first column since there are two zeros, so only one 2×2 determinant, the minor of entry $5 - \lambda$, needs to be computed (Singh, 2015, pp.439-441, 497-498):

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (5-\lambda) \det \begin{pmatrix} 2-\lambda & 1 \\ 0 & -1-\lambda \end{pmatrix} = (5-\lambda)((2-\lambda)(-1-\lambda) - (1)(0)) = (5-\lambda)(2-\lambda)(-1-\lambda)$$

Now solving where $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$:

$$(5-\lambda)(2-\lambda)(-1-\lambda) = 0$$

which gives the three eigenvalues $\lambda_1 = 5$, $\lambda_2 = 2$ and $\lambda_3 = -1$.

The above eigenvalues are the entries on the diagonal of $\mathbf{A}$ which confirms what Singh states, that the eigenvalues for a triangular matrix like $\mathbf{A}$ are the values on the leading diagonal (Singh, 2015, p.507).

Now we need to substitute each eigenvalue into $(\mathbf{A} - \lambda \mathbf{I})\vec{u} = \vec{0}$ and solve for each $\vec{u}$.

$\lambda_1 = 5$:

$$(\mathbf{A} - 5\mathbf{I})\vec{u}_1 = \vec{0}$$

$$\begin{pmatrix} 0 & 1 & -3 \\ 0 & -3 & 1 \\ 0 & 0 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Starting with row 3: $-6z = 0$ so $z = 0$. The second row is $-3y + z = 0$ so if we substitute in $z = 0$ we get $y = 0$. The first column is all zeros so x can be any value, it is a free variable. Let $x = t$ where $t \neq 0$.

$$u_1 = \begin{pmatrix} t \\ 0 \\ 0 \end{pmatrix} = t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$\lambda_2 = 2$:

$$\begin{pmatrix} 3 & 1 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

From row two, $z = 0$. If we substitute this into row one we get $3x + y - 3(0) = 0$ which simplifies to $y = -3x$. Let $x = t$ where $t \neq 0$:

$$u_2 = \begin{pmatrix} t \\ -3t \\ 0 \end{pmatrix} = t \begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix}$$

$\lambda_3 = -1$:

$$\begin{pmatrix} 6 & 1 & -3 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The second row simplifies to $z = -3y$ and substituting this into row one results in $6x + y - 3(-3y) = 0$ which simplifies to $x = -\frac{5}{3}y$. Let $y = -3t$ where $t \neq 0$, then $z = 9t$ and $x = 5t$. So:

$$u_3 = \begin{pmatrix} 5t \\ -3t \\ 9t \end{pmatrix} = t \begin{pmatrix} 5 \\ -3 \\ 9 \end{pmatrix}$$

In conclusion the eigenvalues and eigenvectors are:

$$\lambda_1 = 5 \text{ and } u_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \lambda_2 = 2 \text{ and } u_2 = \begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix}, \lambda_3 = -1 \text{ and } u_3 = \begin{pmatrix} 5 \\ -3 \\ 9 \end{pmatrix}$$

3. From your own research, explain how triangular matrices are used in graphics to capture scale and shear information (the 2D case suffices).

Answer:

As displayed in Vince (2025, pp.200-201) a 2D shear is an upper triangular matrix (as defined in question 8) with the shear factor stored in the off-diagonal:

$$A = \begin{pmatrix} 1 & \tan \beta \\ 0 & 1 \end{pmatrix}$$

Vince (2025, pp.198-199) also shares a matrix for scaling by s_x, s_y which has the scale factors on the diagonal, as can be seen below:

$$B = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$$

Note that I have ignored his homogeneous matrix representation for the purposes of this question since the focus is the 2×2 matrix.

I was then left with the question of why this is relevant to graphics and I thought about what I do when I edit my own images; I apply multiple edits and that is equivalent to multiplying matrices together. Below I provide an example of this where I multiply ABA so reading right-to-left, applying a shear, then scaling and then a shear again:

$$A \cdot B \cdot A = \begin{pmatrix} 1 & \tan \beta \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix} \cdot \begin{pmatrix} 1 & \tan \beta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_x & (s_x + s_y) \tan \beta \\ 0 & s_y \end{pmatrix}$$

The result of the matrix multiplication was interesting to me, as you are left with an upper triangular matrix. I was able to confirm from Theorem 1.7.1(b) in Anton and Kaul (2019, p.71), that the product of upper triangular matrices is upper triangular. This result also highlights how the two operations continue to live in separate compartments, the scaling effects are localised to the diagonal while the shear is in the off-diagonal, even after multiple transformations.

The determinant is then $s_x \cdot s_y$ which shows that the scaling of the area is only determined by

the diagonal and not by the angle of shear. This makes sense when considering just the 2×2 shear matrix whose determinant is 1 so no matter what the angle of shear is, the image area is unaffected. I know when I am editing images I like to be able to undo operations which would then be as simple as calculating the inverse $[ABA]^{-1}$ which is easier for an upper triangular matrix given the zeros under the diagonal.

I did some additional research as I was curious about why this is useful in graphics and I found Paeth (1990, p.189) who explained that all these intermediate operations can be accumulated into a single composite matrix that is then applied to the image once and in doing so avoids unnecessary operations on the image which saves time.

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